

Customer Shopping Behavior Analysis

Turning 3,900 retail transactions into pricing, loyalty and marketing decisions.

\$233,081

TOTAL REVENUE

3,900

TRANSACTIONS

\$59.76

AVG. ORDER VALUE

3.75 / 5

AVG. REVIEW

INSIDE THE REPORT

- 01 The business problem and how the data was prepared
- 02 What the data actually says: revenue, discounts, loyalty
- 03 Product mix, payments and geography
- 04 Five recommendations, with the evidence behind each

Tools · Python (pandas) · MySQL · Power BI

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01 · Context

The Business Problem

A retail company selling clothing, footwear, accessories and outerwear across 50 US states had noticed its purchasing patterns shifting across demographics, categories and channels. Leadership could see revenue in aggregate, but not why it moved: which segments actually drove it, whether discounts and subscriptions were paying for themselves, and where marketing spend was being wasted.

The guiding question: “How can the company use consumer shopping data to identify trends, improve customer engagement, and optimise marketing and product strategy?”

3,900 TRANSACTIONS	18 RAW COLUMNS	50 US STATES	25 PRODUCTS
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Scope of Work

- Python (pandas) — cleaning, missing-value treatment, feature engineering and export of an analysis-ready table.
- MySQL — a structured customer table and ten business queries covering revenue, loyalty, discounts and product mix.
- Power BI — an interactive one-page dashboard with KPI cards and slicers for gender, category, age group and subscription.
- Report & presentation — findings translated into recommendations a marketing team can act on.

Data Preparation

The raw export carried 3,900 rows and 18 columns. Three issues needed fixing before any analysis could be trusted:

Issue found	Treatment applied	Why it matters
37 missing review ratings (0.9% of rows)	Filled with the median rating of the same product category	Keeps every transaction in the revenue totals without distorting category-level satisfaction
Promo Code Used perfectly duplicated Discount Applied	Verified equality across all rows, then dropped the column	Removes a redundant field that would double-count the same behaviour
Inconsistent column names and free-text purchase frequency	Lower-cased snake_case names; mapped frequency to a numeric day interval	Makes the table SQL- and Power BI-friendly and the frequency field sortable

Two engineered fields carry most of the analytical weight: age_group, created with quartile binning into Young Adult, Adult, Middle-aged and Senior, and purchase_frequency_days, which converts labels like “Fortnightly” into 14 days so buying cadence can be compared numerically.

02 · Findings

What the Data Actually Says

Revenue is broad, not concentrated

Total revenue of \$233,081 comes from an average order of \$59.76. Clothing contributes \$104,264 (44.7%) on 1,737 orders, Accessories \$74,200, Footwear \$36,093 and Outerwear \$18,524. Average order value, however, barely moves between categories — \$57.17 for Outerwear to \$60.26 for Footwear. Category revenue is a function of order volume, not basket size.

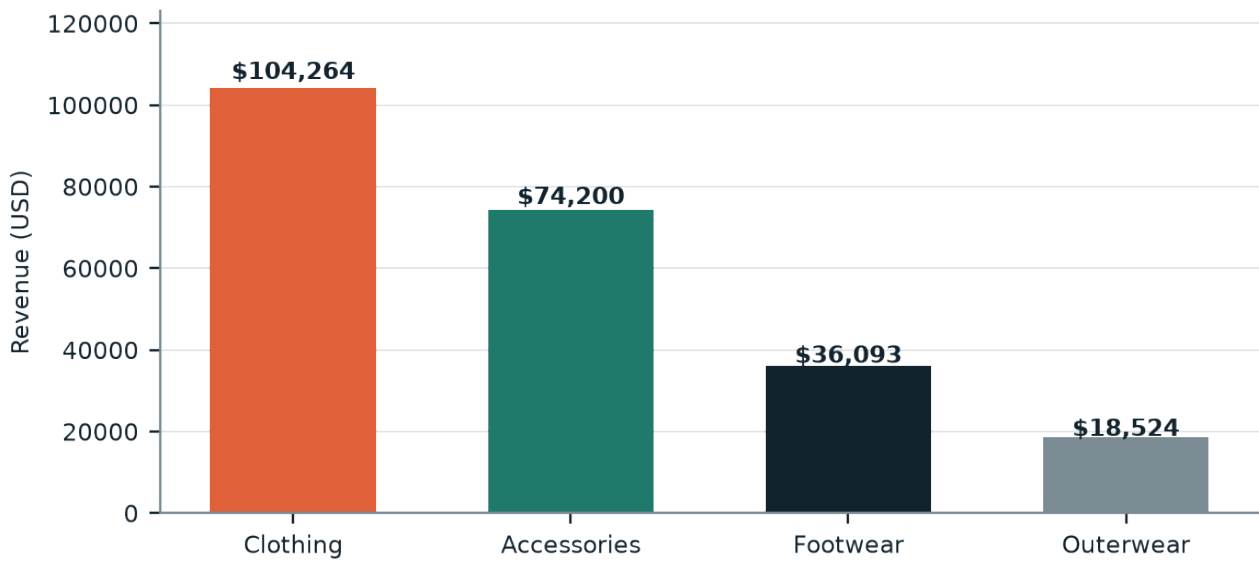


Figure 1 — Revenue by product category. Clothing leads on volume; ticket sizes are near-identical across categories.

No single age group or season carries the business

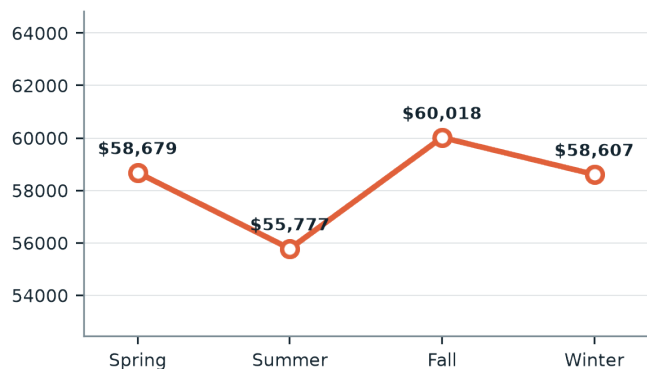
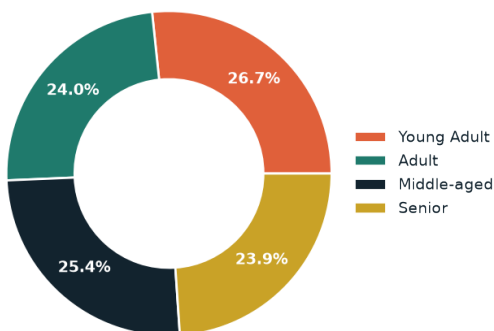


Figure 2 — Revenue share by age quartile (left) and revenue by season (right).

Revenue splits almost evenly across the four age quartiles — Young Adult 26.7%, Middle-aged 25.4%, Adult 24.0%, Senior 23.9% — and across seasons, from \$55,777 in Summer to \$60,018 in Fall, a spread of only 7.6%. There is no dominant demographic to build a campaign around and no dead season to discount away; the customer base is genuinely broad.

02 · Findings

Discounts, Loyalty and Delivery

Discounting is not buying bigger baskets

1,677 of 3,900 orders (43%) had a discount applied. Those orders average \$59.28 against \$60.13 for full-price orders — discounted customers spend slightly less, not more. The programme is transferring margin without lifting order value.

The pattern is also structurally odd: every discounted order and every active subscription in this dataset belongs to a male customer. 63.2% of male customers received a discount and 39.7% hold a subscription, while the figure for female customers is 0% on both. Female customers still generate \$75,191 of revenue at a marginally higher average order (\$60.25 vs. \$59.54) — entirely at full price.

43% ORDERS DISCOUNTED	-\$0.85 AOV VS. FULL PRICE	0% FEMALE CUSTOMERS DISCOUNTED	27% SUBSCRIPTION RATE
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Loyalty is deep but disconnected from subscription

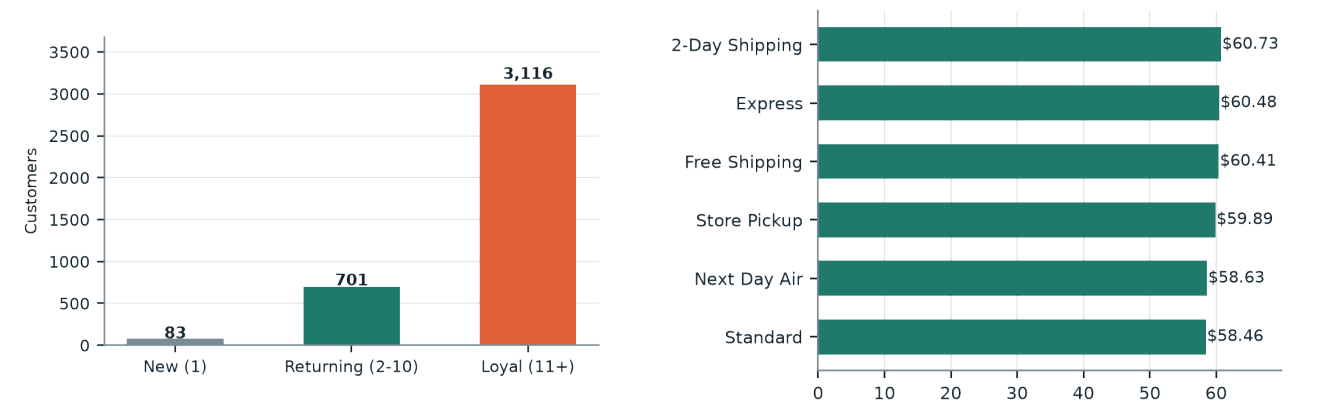


Figure 3 — Loyalty segments by previous-purchase count (left) and average order value by shipping type (right).

Segmenting on previous purchases gives 3,116 Loyal customers (11+ prior orders, 79.9% of the base), 701 Returning and only 83 New. Yet among repeat buyers with more than five prior purchases, just 27.6% are subscribers. The company has already earned the repeat behaviour a subscription is supposed to create — and is failing to convert it.

Shipping choice shows the same flatness: average order runs from \$58.46 on Standard to \$60.73 on 2-Day Shipping, a \$2.27 spread across six options. Premium delivery is not a proxy for a premium customer.

03 · Product mix

Products, Payments and Places

Top item by revenue	Revenue	Top item by rating	Avg. rating
Blouse	\$10,410	Gloves	3.86
Shirt	\$10,332	Sandals	3.84
Dress	\$10,320	Boots	3.82
Pants	\$10,090	Hat	3.80
Jewelry	\$10,010	Handbag	3.78

The top five products by revenue sit within \$400 of each other, and the highest-rated items — Gloves, Sandals, Boots — are none of them. Satisfaction and sales volume are driven by different products, which is exactly the gap a cross-sell programme should exploit: high-rated accessories are natural attachments to high-volume apparel.

Payment methods are similarly even, with Credit Card (\$40,310), PayPal (\$40,109) and Cash (\$40,002) within 1% of each other across six options — there is no payment channel worth prioritising or retiring. Geographically, Montana (\$5,784), Illinois (\$5,617) and California (\$5,605) top the table, but the leading state accounts for only 2.5% of revenue.



Figure 4 — Revenue split by gender. Female customers contribute 32.3% of revenue entirely at full price.

The overall satisfaction picture is stable and mid-range: an average rating of 3.75 / 5, with category medians used to repair the 37 gaps. No category rates poorly enough to signal a quality problem, and none rates highly enough to be used as a marketing claim.

04 · Recommendations

What To Do About It

Recommendation	The evidence behind it	Expected effect
Cap and target the discount programme. Restrict discounts to reactivation and first-order offers instead of blanket application.	Discounted orders average \$59.28 vs. \$60.13 at full price across 1,677 discounted orders.	Recovers margin on ~43% of orders with no expected drop in order value.
Convert loyal buyers into subscribers. Trigger a subscription offer once a customer passes five orders.	Only 27.6% of repeat buyers (5+ prior purchases) currently subscribe, despite 3,116 Loyal customers.	Turns proven repeat behaviour into predictable recurring revenue.
Open the discount and subscription funnel to female customers.	0% of female customers hold a discount or subscription, yet they generate \$75,191 at a higher AOV (\$60.25).	Unlocks an untouched segment already buying at full price.
Bundle high-rated accessories with high-volume apparel.	Top-rated items (Gloves 3.86, Sandals 3.84) are absent from the top five by revenue (Blouse, Shirt, Dress).	Raises basket size, the metric discounts have failed to move.
Stop segmenting campaigns by season, age or payment method. Segment on behaviour instead.	Seasonal revenue varies only 7.6%; age quartile shares sit between 23.9% and 26.7%; payment methods within 1%.	Redirects spend from cosmetic segments to ones that predict spend.

Limitations

The dataset is a single cross-sectional snapshot: it records prior-purchase counts but no order dates, so true cohort retention, churn timing and customer lifetime value cannot be calculated. The perfect gender split on discounts and subscriptions is strong enough to be worth validating against the source system before acting on it. Adding transaction timestamps and acquisition channel would make the next iteration materially stronger.

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